

Lab Assignment 8: Anomaly Detection and Physiological Signal Classification

CSAI2017P: Applied Machine Learning (Lab)

Experiments 10–11 (Units VI–VII) • CO2, CO3, CO4

Instructor: Dr. Tofik Ali

Due: announced in the lab

Student Name: _____

SAP ID: _____

Batch: _____

Lab quiz: LQ8, after submission

Objective

- Detect outliers without labels and judge the result honestly.
- Classify windows of sensor data from engineered features.
- Respect subject-wise splits when rows are not independent.

Datasets used in this assignment

A2 (Telco), C3 (UCI Human Activity Recognition).

Full descriptions, sources and loading snippets for all anchor datasets are in **Anchor-Datasets.pdf**, alongside this brief.

Instructions

- Use a train/test split (80/20 unless stated) and set `random_state` for reproducibility.
- Fit every transformer inside a `Pipeline` so nothing is fitted on the test set.
- Report the metric named in the question. A number without units or context earns no marks.
- Every question asks for a short written observation. Code without commentary caps you at half marks.
- The lab quiz that follows will ask you to read, trace and modify *your own* submitted code.

Submit:

- Notebook: `AML_LA08_<SAPID>.ipynb` with all cells executed and outputs visible.
- Report: 2–3 pages PDF, or a `README.md`, carrying your results table and observations.
- Do not upload datasets. Cite the source and include the loading code.

Section	Level	Choice rule	Marks
A	Easy	Attempt any 2 of 3 (2 marks each)	4
B	Medium	Attempt any 1 of 2 (4 marks each)	4
C	Hard	Compulsory	2
Total			10

Course outcomes assessed by this assignment: **CO2, CO3, CO4**. Attempting more than the choice rule allows is not penalised; the best attempts are marked.

Section A (Easy) — attempt any 2 of 3 ($2 \times 2 = 4$)**A1. Statistical outliers****[2 marks]***Dataset: A2, numeric columns.*

- (a) Flag outliers in `tenure`, `MonthlyCharges` and `TotalCharges` using the IQR rule and the 3σ rule.
- (b) Report how many rows each rule flags and how many they agree on.
- (c) State in two lines which rule you would trust on a skewed column, and why.

A2. Isolation Forest**[2 marks]***Dataset: A2.*

- (a) Fit `IsolationForest` with `contamination=0.05` on the scaled numeric features.
- (b) Report how many rows are flagged and inspect five of them.
- (c) State in two lines what makes those rows unusual.

A3. Load the signal features**[2 marks]***Dataset: C3 (HAR).*

- (a) Load the pre-computed 561-feature train and test sets and report both shapes.
- (b) Report the class distribution across the six activities.
- (c) State in two lines why the authors supplied their own split rather than letting you make one.

Section B (Medium) — attempt any 1 of 2 ($1 \times 4 = 4$)**B1. Subject-wise versus random split****[4 marks]***Dataset: C3 (HAR).*

- (a) Train a linear SVM using the supplied subject-wise split and report test accuracy.
- (b) Pool train and test, reshuffle randomly at the same ratio, retrain and report accuracy again.
- (c) Explain in 6–8 lines why the random figure is inflated and what it would cost you in deployment.

B2. Anomalies without labels**[4 marks]***Dataset: A2 or C3, your choice.*

- (a) Run Isolation Forest, One-Class SVM and Local Outlier Factor at the same contamination rate.
- (b) Report how many points all three agree on, and how many only one method flags.
- (c) In 6–8 lines, explain how you would decide which method is right when you have no labels.

Section C (Hard) — compulsory (2)

C1. Which activities blur

[2 marks]

Dataset: C3 (HAR).

- (a) Plot the confusion matrix from your subject-wise model.
- (b) Name the two activities most often confused and explain in 4–6 lines why their sensor signatures resemble each other.

End of Lab Assignment 8

Sources: Müller and Guido, *Introduction to Machine Learning with Python*, companion notebooks.;McKinney, *Python for Data Analysis* (3e), O'Reilly, 2022.;Scikit-learn user guide, accessed 2026-08-04.;Matplotlib and Seaborn documentation, accessed 2026-08-04.